Status of the Gunter & Scott 1990 Development in Lean

Measured on `main` at `5a1d8bc`. Build 1372 jobs, 0 errors, 0 warnings. 124 modules, 46,097 lines, 2,119 theorems.

1. Extractions — what the paper contains

#	Kind	Count
1	Definitions	≈13
2	Numbered theorems	16
3	Numbered lemmas	13
4	Numbered propositions	1
5	— numbered results, total	**30**
6	— the conjuncts those 30 decompose into	93
7	Prose statements	146
8	Examples	**not measured**

Numbers 1–30 are contiguous, no gap and no repeat, verified by extracting every `(Theorem\|Lemma\|Proposition\|Corollary) N` heading from the full text.

Two rows are soft. Row 1 has stood at "≈13" since r0031 and has never been enumerated to an exact figure. Row 8 is a real hole: no round has counted the paper's examples at all. Figure 4's element counts (1, 2, 5, 20) were used to adjudicate a §7.4 reading, so at least some examples have been read — but never enumerated. Both should be counted the same way the 30 were: extract the headings from the full text.

2. Represented in Lean

#	Kind	Represented	Of	Not stated
1	Definitions	≈13	≈13	0
2	Theorems	16	16	0
3	Lemmas	13	13	0
4	Propositions	1	1	0
5	Prose statements	86	146	60
6	Examples	—	—	not measured

All 30 numbered results have been stated in Lean since r0048. The 60 unstated prose statements are concentrated rather than spread: 13 of the 26 unstated *properties* are §7 alone.

3. Corrected — a statement that could not be transcribed as printed

#	Kind	Printing error	Reading error	Semantic error	Total
1	Definitions	1	0	1	2
2	Theorems	1	1	2	4
3	Lemmas	1	2	0	3
4	Propositions	0	0	0	0
5	Prose statements	—	—	—	not classified
6	Examples	—	—	—	not measured

Column meanings, and every row behind the counts:

Printing error — the printed page is unreadable or garbled; the mathematics is right.

#	Where	What
1	Lemma 9	the PDF drops every `⊗` and every `⊥`, so which operators the laws range over is unreadable
2	Theorem 14	the list of characterizations is garbled
3	§7.4's worked example	reverses the paper's own definition — printed `b ⊢ a` where both papers give `a ⊢ b`

Reading error — *ours*. The paper is right; we transcribed something it does not say.

#	Where	What
1	Theorem 29, sentence 2	we dropped the word "domain" from "any bifinite domain", which makes the sentence false
2	Lemma 28	we quantified over all `U`; the paper states it over its own `U`
3	Lemma 30	same shape — we wrote a universal closure the paper does not claim

Semantic error — the mathematics is wrong as stated.

#	Where	Whose	What
1	Theorem 26	**the paper's**	false for any signature with two or more 0-ary slots, including the paper's own worked `(2,0,0,0,0,0)`
2	Theorem 7	ours	our guard tested compactness where the sentence is about boundedness; on the paper's own index the join always exists, so it tests neither
3	§7.4's order relation	**the paper's**	the printed relation is not reflexive — `b = (⊥,⌀)` fails `b ⊢ b`. It is the strict part, and the order is its reflexive closure, which is the identification §7.4 then performs by hand

Lemma 9's two misprinted items (3 and 5) are also individually false as printed; they are counted once under Lemma 9's printing error, since a single unreadable page produced both.

4. Proof state

#	Kind	Proven	Unproven	Of
1	Definitions	n/a	n/a	—
2	Theorems	14	2	16
3	Lemmas	12	1	13
4	Propositions	1	0	1
5	— numbered results	**27**	**3**	30
6	Prose statements	70	16	86 stated
7	Examples	—	—	not measured

A definition is not proven or unproven; the thing that can go wrong is that it is **empty** — a definition nothing satisfies makes every theorem over it vacuous. The measurement is instantiation: **25 structures and classes, 0 never instantiated.** Two mechanisms of vacuity have been found and are now checked for: fields inhabited free by `Classical.dec`, and a conclusion parameter with no hypothesis parameter in its component. The second sweep returned 1 hit in 2,111 declarations, the already-known one.

The three unproven numbered results, and what each still needs:

#	Result	State
1	Theorem 7	sentences 1 and 2 proven; the recursive forms are not
2	Theorem 29	sentence 1 proven; sentence 2 is not
3	Lemma 30	stated at `V`; arity 1 — one named input left

All three reduce to two statements, and nothing else is outstanding:

#	The unproven statement	Blocks
1	`RecursiveNormal` for `K(D → E)` — decide whether two basis elements are bounded, and compute where their join is	Theorem 7
2	`Thm29Normal` at infinite `K(E)` — realize the type over the tower's image, not the η-image. Finite bases are proven	Theorems 29, Lemma 30

Neither is unsolved mathematics: Gunter & Scott proved both in 1990.

5. Proof work state

#	Kind	Started, left with a `sorry`
1	Definitions	0
2	Theorems	0
3	Lemmas	0
4	Propositions	0
5	Prose statements	0
6	Examples	0

`sorry` is 0 across the package, and has been since r0042. So is `axiom`, and so is any constant naming `sorryAx`.

That number does not mean everything is proven, and the difference matters when reading table 4. A `sorry` is a hole in a proof somebody *started*. The three unproven results were not started and left open — each is carried as an explicit hypothesis, so a theorem that needs one takes it as an argument. Nothing is asserted that is not proven, and no `sorryAx` propagates; but no `sorry` count, here or anywhere, can see this work.

Reproducing

```
scripts/counts.sh                    # modules, lines, theorems, sorry
scripts/numbered-status.sh           # declarations per numbered result
scripts/a6-env-scan.sh <out>         # then a6-summarize.py
scripts/a8-claim-check.sh            # prose staleness against the environment
scripts/compile.sh -r <round>        # build
```